# PRACTICAL 1B

**Aim: Write a Java Program to to make a class called as circle. It should have three methods namely: accept radius, calculate area, and display the area.**

## Input:

```
import java.util.Scanner;

class Circle{
   double radius, area;

   void acceptRadius(){
      Scanner sc=new Scanner(System.in);
      System.out.print("Enter radius: ");
      radius= sc.nextDouble();
   }

   void calculateArea(){
      area=3.14*radius*radius;
   }

   void displayArea(){
      System.out.println("Area of circle= "+ area);
   }
}

public class CircleDemo{
   public static void main(String[] args){
      Circle c=new Circle();
      c.acceptRadius();
      c.calculateArea();
      c.displayArea();
   }
}
```

## Output:

```
Enter Radius: 5
Area = 78.5
```

# PRACTICAL 2A

**Aim: Write a Java program to demonstrate use of Access specifiers.**

**Input:**

```
class AccessSpecifier {
   private String a=("Member 1");
   String b=("Member 2");
   protected String c=("Member 3");
   public String d=("Member 4");

   // This will help to print a
   public void printA(){
      System.out.println(a+ " is Private Member");
   }
}

public class AccessExample {
   public static void main(String[] args){
      Access obj=new Access();

      obj.printA();

      System.out.println(obj.b+ " is Default Member");
      System.out.println(obj.c+ " is Protected Member");
      System.out.println(obj.d+ " is Public Member");
   }
}
```

**Output:**

```
Member 1 is Private Member
Member 2 is Default Member
Member 3 is Protected Member
Member 4 is Public Member
```

# PRACTICAL 3B

**Aim: Write a Java Program to perform division of two user-entered numbers and handle NumberFormatException and ArithmeticException using try-catch blocks.**

## Input:

```
import java.util.Scanner;

public class DivisionExample {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      try {
         // input numbers as strings
         System.out.print("Enter first number: ");
         String str1 = sc.next();
         System.out.print("Enter second number: ");
         String str2 = sc.next();

         // may throw NumberFormatException
         int num1 = Integer.parseInt(str1);
         int num2 = Integer.parseInt(str2);

         // may throw ArithmeticException
         int result = num1 / num2;

         System.out.println("Result = " + result);

      } catch (NumberFormatException e) {
         System.out.println("Error: Please enter valid numbers!");
      } catch (ArithmeticException e) {
         System.out.println("Error: Division by zero is not allowed!");
      }
   }
}
```

# PRACTICAL 4A

**Aim: Write two Java programs to print numbers from 1 to 10 using:**

1. **By implementing the Runnable interface.**
2. **By extending the Thread class.**

**Input: By implementing the Runnable interface.**

```
class Numbers implements Runnable {
   public void run() {
      for (int i = 1; i <= 10; i++) {
         System.out.println(i);
      }
   }
}

public class Main {
   public static void main(String[] args) {
      Numbers n = new Numbers();
      Thread t1 = new Thread(n);
      t1.start();
   }
}
```

**Output:**

```
1
2
3
4
5
6
7
8
9
10
```

# PRACTICAL 4B

**Aim: Write Java programs to print even and odd numbers from 1 to 50 using:**

1. **Runnable interface.**
2. **Extending the Thread class.**

**Input: By implementing the Runnable interface.**

```
class Numbers implements Runnable{
   public void run() {
      for (int i=1; i<=50; i++){
         if (i%2==0) {
            System.out.println (i + ":is an even number.");
         } else
         {
            System.out.println (i + ":is an odd number.");
         }
      }
   }
}

public class Main {
   public static void main(String[] args) {
      Numbers n= new Numbers();
      Thread t1= new Thread(n);
      t1.start();
   }
}
```

# PRACTICAL 5A

**Aim: Write a Java program to demonstrate the use of methods in linked list instance.**

**Input:**

```
import java.util.*;

class LinkedListDemo {
   public static void main(String ar[]) {
      LinkedList<String> list = new LinkedList<>();

      list.add("One");
      list.add("Two");
      list.add("Three");
      list.add("Four");
      list.add("Five");
      list.addLast("Six");
      list.addFirst("Seven");
      list.add(1, "Eight");

      System.out.println("Original Contents of list: " + list);

      list.remove("Five");
      list.remove(2);
      System.out.println("\nContents of list after deletion: " + list);

      list.removeFirst();
      list.removeLast();
      System.out.println("\nList after deleting first and last: " + list);

      String val = list.get(2);
      list.set(2, val+ "-Changed");
      System.out.println("\nList after change: " + list);
   }
```

# PRACTICAL 9

**Aim: Write a java program for JSP code to print the details entered in the form on next page.**

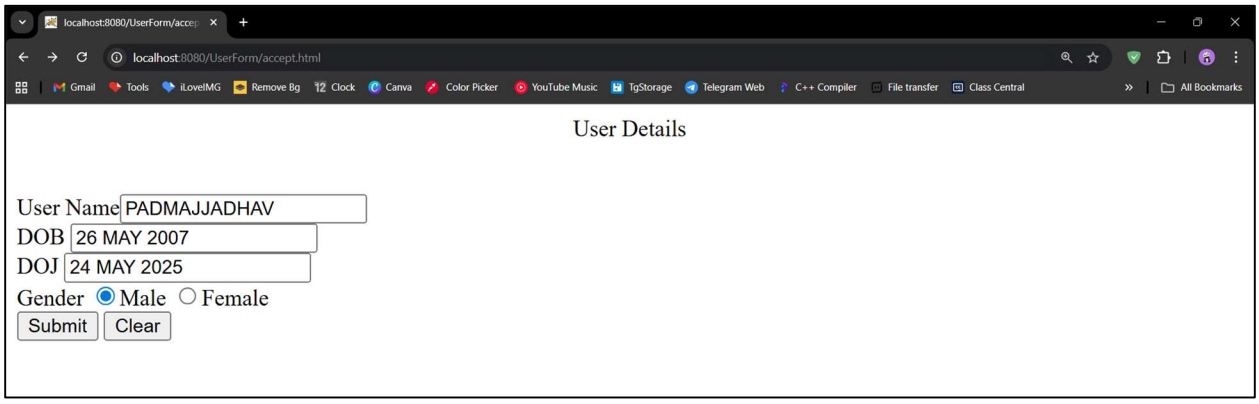
**Input: accept.html**

```
<html>

<body>
   <form name=frm method=post action=display.jsp>
      <center> User Details</center>
      <br><br> User Name<input type=text name=t1>
      <br> DOB <input type=text name=t2>
      <br> DOJ <input type=text name=t3>
      <br> Gender
      <input type=radio name=r1 value="Male">Male
      <input type=radio name=r1 value="Female">Female
      <br><input type=submit value=submit>
      <input type=reset value=clear>
   </form>
</body>

</html>
```

**Output:**

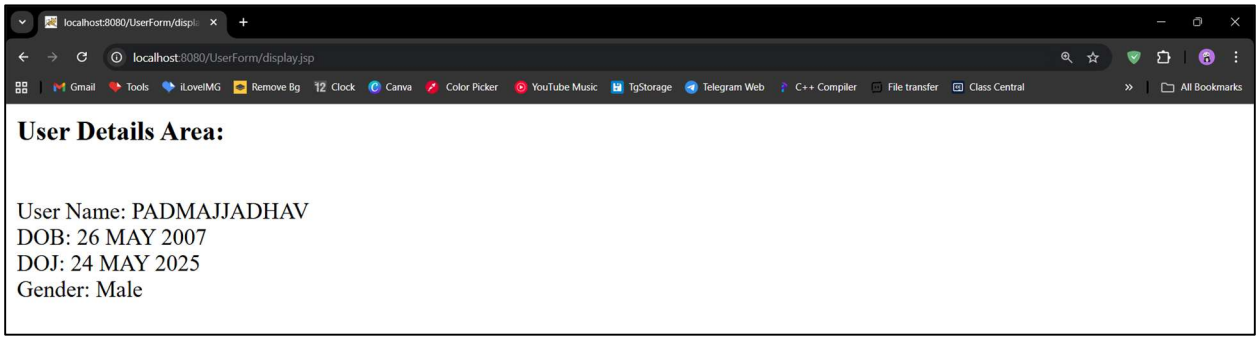


## Input: display.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
pageEncoding="UTF-8"%>
<html>
<body>
<%
   String username = request.getParameter("t1");
   String dob      = request.getParameter("t2");
   String doj      = request.getParameter("t3");
   String gender   = request.getParameter("r1");

   out.println("<h3>User Details Area:</h3>");
   out.println("<br>User Name: " + username);
   out.println("<br>DOB: " + dob);
   out.println("<br>DOJ: " + doj);
   out.println("<br>Gender: " + gender);
%>
</body>
</html>
```

## Output:



# PRACTICAL 10

**Aim: Write a java program for JavaScript file with JSON.**

**Input:**

```
<!DOCTYPE html>
<html>
<head>
   <title>First JSON example</title>
</head>
<body>
   <div id="content"></div>

   <script type="text/javascript">
      var object1 = { "name": "Mango", "rate": 250 };
      var object2 = { "name": "Apple", "rate": 150 };

      var content = "<h2>JSON with JavaScript example</h2><br>";

      content += "<h3>Fruit Name = " + object1.name + "</h3>";
      content += "<h3>Rate/Kg = " + object1.rate + "</h3><br>";

      content += "<h3>Fruit Name = " + object2.name + "</h3>";
      content += "<h3>Rate/Kg = " + object2.rate + "</h3><hr>";

      content += object1.name + " is available at " + object1.rate + " rate/kg<hr>";
      content += object2.name + " is available at " + object2.rate + " rate/kg<hr>";

      document.getElementById("content").innerHTML = content;
   </script>
</body>
</html>
```